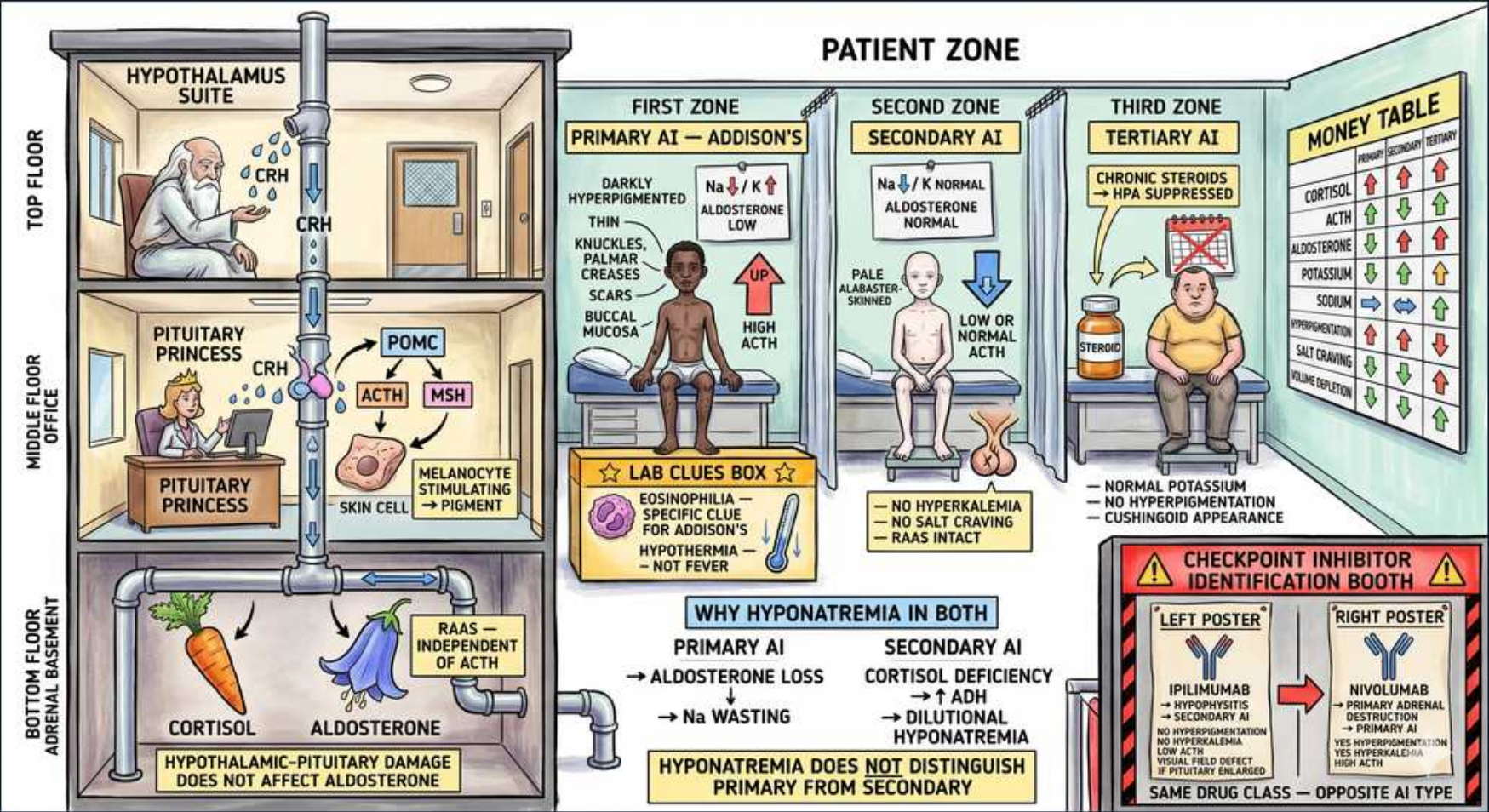


Adrenal Insufficiency/CAH — Three-Bucket Hospital (Primary vs Secondary vs Tertiary)



1. Hypothalamus / CRH

WHAT YOU SEE

Top-floor hypothalamus suite, king releasing CRH

WHAT IT ENCODES

Top floor of the axis: the hypothalamus secretes CRH, peaking in the early morning. Failure here — or suppression by chronic exogenous glucocorticoids — produces TERTIARY AI. Because the defect is above the pituitary and adrenal, ACTH is low/normal and aldosterone is preserved (RAAS intact). CRH is the trigger that the cosyntropin test bypasses by giving synthetic ACTH directly.

BOARD TRAP

Tertiary AI (chronic steroid suppression) is the most common cause of AI overall, and abrupt steroid withdrawal precipitates crisis.

2. Pituitary ACTH + POMC

WHAT YOU SEE

Middle floor pituitary princess making POMC → ACTH + MSH

WHAT IT ENCODES

Middle floor: the pituitary cleaves POMC into ACTH and MSH (melanocyte-stimulating hormone). ACTH drives cortisol and adrenal androgens. In PRIMARY AI, lost cortisol feedback drives ACTH sky-high, and the co-secreted MSH stimulates melanocytes → hyperpigmentation. In secondary AI the corticotrophs fail, so both ACTH and MSH are low → pale skin.

BOARD TRAP

Hyperpigmentation is an ACTH/MSH (POMC) phenomenon — present in primary AI (high ACTH), absent in secondary/tertiary (low ACTH).

3. Cortisol + aldosterone basement

WHAT YOU SEE

Bottom floor adrenal basement: cortisol carrot + aldosterone (RAAS) independent of ACTH

WHAT IT ENCODES

Basement (adrenal cortex) makes cortisol under ACTH control and aldosterone under RAAS/potassium control. This split is the key to the whole topic: any lesion in the hypothalamus or pituitary cuts cortisol but leaves aldosterone intact, whereas destruction of the adrenal itself kills BOTH. Androgens (DHEA-S) track with cortisol/ACTH.

BOARD TRAP

Aldosterone is RAAS-controlled, so pituitary/hypothalamic disease spares it — only adrenal (primary) disease causes mineralocorticoid loss.

4. Primary AI — Addison's

WHAT YOU SEE

First zone PRIMARY AI ADDISON'S, darkly hyperpigmented patient

WHAT IT ENCODES

Primary AI (Addison disease) is destruction of the adrenal cortex, so ALL three outputs fall: LOW cortisol, LOW aldosterone, low DHEA-S, with HIGH ACTH and HIGH renin. The clinical picture is fatigue, weight loss, hypotension/orthostasis, salt craving, hyperpigmentation, plus hyponatremia and hyperkalemia. Confirm with 8 AM cortisol, cosyntropin (peak <18), high ACTH, low aldosterone/high renin, and 21-hydroxylase antibodies.

BOARD TRAP

Only PRIMARY AI combines high ACTH with low aldosterone (and therefore hyperkalemia + hyperpigmentation).

5. High ACTH arrow

WHAT YOU SEE

Big red HIGH ACTH arrow rising in the primary zone

WHAT IT ENCODES

With the adrenal destroyed, cortisol can't rise to exert negative feedback, so CRH and ACTH climb to very high levels. Measuring ACTH is what localizes the lesion: HIGH ACTH = primary; LOW/inappropriately-normal ACTH = secondary/tertiary. The high ACTH also explains the hyperpigmentation via MSH and is the basis for the high-baseline picture even before stimulation testing.

BOARD TRAP

It's the ACTH level — not the cortisol level — that distinguishes primary from central AI.

6. Hyperpigmentation

WHAT YOU SEE

Thin, darkly pigmented knuckles/palmar creases/scars/buccal mucosa

WHAT IT ENCODES

Chronically elevated ACTH/MSH stimulates melanocytes, darkening sun-exposed and friction areas, palmar creases, knuckles, recent scars, and the buccal/gingival mucosa. It is a hallmark of PRIMARY AI and a clue to chronicity (takes time to develop). It is absent in central AI because ACTH/MSH are low.

BOARD TRAP

Buccal/palmar-crease/scar hyperpigmentation points specifically to PRIMARY AI; a PALE patient with AI is central.

7. Low Na / high K (primary)

WHAT YOU SEE

Lab box Na down / K up in the primary zone

WHAT IT ENCODES

Aldosterone loss in primary AI removes distal Na⁺ reabsorption and K⁺/H⁺ excretion → renal salt wasting with hyponatremia, HYPERkalemia, mild non-anion-gap metabolic acidosis, salt craving, and volume depletion/orthostatic hypotension. Cortisol deficiency adds dilutional hyponatremia via elevated ADH. Hypoglycemia and eosinophilia/mild hypercalcemia can also appear.

BOARD TRAP

HYPERKALEMIA is the discriminator — its presence indicates aldosterone loss and thus PRIMARY (not secondary) AI.

8. Secondary AI

WHAT YOU SEE

Second zone SECONDARY AI, pale undersized patient

WHAT IT ENCODES

Secondary AI is pituitary ACTH deficiency (tumor, surgery, Sheehan, apoplexy, hypophysitis): LOW cortisol, LOW/normal ACTH, but PRESERVED aldosterone (RAAS intact). Skin is pale (low MSH), there is no salt craving or hyperkalemia, and patients may have other pituitary hormone deficits. Hyponatremia can still occur from cortisol-deficiency-driven ADH excess.

BOARD TRAP

Secondary AI = pale skin, normal potassium, intact aldosterone — the mirror image of Addison's pigmented/hyperkalemic picture.

9. Aldosterone normal (secondary)

WHAT YOU SEE

Secondary zone note Na^+/K^+ normal, aldosterone normal

WHAT IT ENCODES

Because ACTH does not control the glomerulosa, aldosterone stays normal in secondary (and tertiary) AI: potassium is normal, there is no salt craving, and blood pressure depletion is milder. Any hyponatremia is dilutional (cortisol deficiency → ADH), not from sodium wasting. Therefore central AI is treated with glucocorticoid ALONE — no fludrocortisone needed.

BOARD TRAP

Don't give fludrocortisone in central AI — aldosterone is already intact; glucocorticoid replacement is sufficient.

10. Tertiary AI / chronic steroids

WHAT YOU SEE

Third zone TERTIARY AI, chronic steroids, HPA suppressed, moon/withdrawal

WHAT IT ENCODES

Tertiary AI is hypothalamic CRH deficiency, most often from chronic exogenous glucocorticoids suppressing the HPA axis (and causing adrenal atrophy). It is the most common cause of AI overall. Patients may look Cushingoid from the steroids yet be functionally adrenally insufficient if the drug is stopped abruptly or a stressor hits. Manage by tapering and stress-dosing.

BOARD TRAP

Never stop chronic steroids abruptly — taper slowly and stress-dose for illness/surgery to avoid precipitating adrenal crisis.

11. Money table

WHAT YOU SEE

MONEY TABLE comparing cortisol/ACTH/aldosterone/K/Na across the three types

WHAT IT ENCODES

The summary table to memorize: PRIMARY → cortisol low, ACTH HIGH, aldosterone LOW, K HIGH, Na low, hyperpigmented, salt craving. SECONDARY/TERTIARY → cortisol low, ACTH LOW/normal, aldosterone NORMAL, K normal, Na low (dilutional), pale, no salt craving. Hyperpigmentation and hyperkalemia are the two findings unique to primary AI.

BOARD TRAP

The only columns that DIFFER between types are ACTH, aldosterone, potassium, pigmentation, and salt craving — cortisol and hyponatremia are low in all.

12. Why hyponatremia in both

WHAT YOU SEE

Bottom panel: why hyponatremia occurs in primary AND secondary

WHAT IT ENCODES

Hyponatremia is the great non-discriminator. In primary AI it comes from aldosterone loss (renal Na wasting) PLUS cortisol-deficiency ADH excess. In secondary AI it is purely from cortisol deficiency: low cortisol fails to suppress ADH → water retention → dilutional hyponatremia. So low sodium appears in both and cannot localize the lesion.

BOARD TRAP

Hyponatremia alone does NOT distinguish primary from secondary AI — use potassium, ACTH, and aldosterone/renin instead.

13. Aldosterone preserved note

WHAT YOU SEE

Caption: hypothalamic-pituitary damage does not affect aldosterone

WHAT IT ENCODES

The take-home caption: hypothalamic or pituitary damage does NOT affect aldosterone because the glomerulosa answers to renin/angiotensin II and potassium, not ACTH. This single fact explains the entire secondary/tertiary profile — normal K^+ , no salt craving, milder volume problems — and why fludrocortisone is unnecessary centrally.

BOARD TRAP

If a patient with AI has hyperkalemia, the lesion CANNOT be purely central — aldosterone is being lost, so it's primary.

14. Checkpoint inhibitor booth

WHAT YOU SEE

Red booth CHECKPOINT INHIBITOR IDENTIFICATION: ipilimumab vs nivolumab

WHAT IT ENCODES

Immune checkpoint inhibitors are a high-yield modern cause of AI. CTLA-4 inhibitors (ipilimumab) classically cause hypophysitis → secondary AI, while PD-1/PD-L1 inhibitors (nivolumab, pembrolizumab) more often cause primary adrenalitis or thyroiditis. Same drug class (immunotherapy) but they can hit opposite levels of the axis — check ACTH to localize.

BOARD TRAP

Checkpoint-inhibitor hypophysitis (ipilimumab) gives LOW ACTH/secondary AI; direct adrenalitis gives HIGH ACTH/primary AI — don't assume the level.